

timers are also simulated, and timer interrupts will cause your timer handling routine to be activated.

The full lab assignment, as well as code you will need to compile with your own code, are available at this book's Web site: <http://www.awl.com/kurose-ross>.



## Wireshark Lab: Exploring TCP

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In this lab, you'll use your Web browser to access a file from a Web server. As in earlier Wireshark labs, you'll use Wireshark to capture the packets arriving at your computer. Unlike earlier labs, you'll *also* be able to download a Wireshark-readable packet trace from the Web server from which you downloaded the file. In this server trace, you'll find the packets that were generated by your own access of the Web server. You'll analyze the client- and server-side traces to explore aspects of TCP. In particular, you'll evaluate the performance of the TCP connection between your computer and the Web server. You'll trace TCP's window behavior, and infer packet loss, retransmission, flow control and congestion control behavior, and estimated roundtrip time.

As is the case with all Wireshark labs, the full description of this lab is available at this book's Web site, <http://www.awl.com/kurose-ross>.



## Wireshark Lab: Exploring UDP

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In this short lab, you'll do a packet capture and analysis of your favorite application that uses UDP (for example, DNS or a multimedia application such as Skype). As we learned in Section 3.3, UDP is a simple, no-frills transport protocol. In this lab, you'll investigate the header fields in the UDP segment as well as the checksum calculation.

As is the case with all Wireshark labs, the full description of this lab is available at this book's Web site, <http://www.awl.com/kurose-ross>.

### Van Jacobson

Van Jacobson is a Research Fellow at PARC. Prior to that, he was co-founder and Chief Scientist of Packet Design. Before that, he was Chief Scientist at Cisco. Before joining Cisco, he was head of the Network Research Group at Lawrence Berkeley National Laboratory and taught at UC Berkeley and Stanford. Van received the ACM SIGCOMM Award in 2001 for outstanding lifetime contribution to the field of communication networks and the IEEE Kobayashi Award in 2002 for “contributing to the understanding of network congestion and developing congestion control mechanisms that enabled the successful scaling of the Internet”. He was elected to the U.S. National Academy of Engineering in 2004.



#### **Please describe one or two of the most exciting projects you have worked on during your career. What were the biggest challenges?**

School teaches us lots of ways to find answers. In every interesting problem I've worked on, the challenge has been finding the right question. When Mike Karels and I started looking at TCP congestion, we spent months staring at protocol and packet traces asking “Why is it failing?”. One day in Mike's office, one of us said “The reason I can't figure out why it fails is because I don't understand how it ever worked to begin with.” That turned out to be the right question and it forced us to figure out the “ack clocking” that makes TCP work. After that, the rest was easy.

#### **More generally, where do you see the future of networking and the Internet?**

For most people, the Web is the Internet. Networking geeks smile politely since we know the Web is an application running over the Internet but what if they're right? The Internet is about enabling conversations between pairs of hosts. The Web is about distributed information production and consumption. “Information propagation” is a very general view of communication of which “pairwise conversation” is a tiny subset. We need to move into the larger tent. Networking today deals with broadcast media (radios, PONs, etc.) by pretending it's a point-to-point wire. That's massively inefficient. Terabits-per-second of data are being exchanged all over the World via thumb drives or smart phones but we don't know how to treat that as “networking”. ISPs are busily setting up caches and CDNs to scalably distribute video and audio. Caching is a necessary part of the solution but there's no part of today's networking—from Information, Queuing or Traffic Theory down to the Internet protocol specs—that tells us how to engineer and deploy it. I think and hope that over the next few years, networking will evolve to embrace the much larger vision of communication that underlies the Web.